A PROJECT

REPORT

ON

"Optimization of Railway Reservation System using Reinforcement Learning"

Submitted in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

In

Computer Engineering

GUIDED BY:
Ms. Archika Jain
(ASSISTANT PROFESSOR)

SUBMITTED BY:
NABH SHARMA
NANDINI SINGHAL
NIHARIKA GUPTA
NISHTHA GORANA

RAJASTHAN TECHNICAL UNIVERSITY,KOTA

DEPARTMENT OF COMPUTER ENGINEERING
POORNIMA COLLEGE OF ENGINEERING
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May -2022

Dr. Mahesh Bundele
B.E., M.E., Ph.D.
Director
Poornima College of Engineering
ISI-6, RIICO Institutional Area
Sitapura, JAIPUR

DEPARTMENT CERTIFICATE

This is to certify that Mr. Nabh Sharma registration no. PCE18CS096, of the POORNIMA COLLEGE OF ENGINEERING Department of Computer Engineering, has submitted this project report entitled "Optimization of Railway Reservation System using Reinforcement Learning" under the supervision of Ms. Archika Jain, working as Asst. Professor in department of Computer Engineering as per the requirements of the Bachelor of Technology program of Poornima College of Engineering, Jaipur.

(Name: Dr. Surendra Kumar Yadav)

Head, Dept of Computer Engineering

(Name: Mr. Vishal Choudhary)

Project Coordinator

Dr. Mahesh Bundele
B.E., M.E., Ph.D.
Director
Poornima College of Engineering
ISI-6, RIICO Institutional Area
Sitapura, JAIPUR

DEPARTMENT CERTIFICATE

This is to certify that Ms. Nandini Singhal, registration no. PCE18CS099, of the POORNIMA COLLEGE OF ENGINEERING Department of Computer Engineering, has submitted this project report entitled "Optimization of Railway Reservation System using Reinforcement Learning" under the supervision of Ms. Archika Jain, working as Asst. Professor in department of Computer Engineering as per the requirements of the Bachelor of Technology program of Poornima College of Engineering, Jaipur.

(Name: Dr. Surendra Kumar Yadav)

Head, Dept of Computer Engineering

(Name: Mr. Vishal Choudhary)

Project Coordinator

Dr. Mahesh Bundele
B.E., M.E., Ph.D.
Director
Poornima College of Engineering
ISI-6, RIICO Institutional Area
Sitapura, JAIPUR

DEPARTMENT CERTIFICATE

This is to certify that Ms. Niharika Gupta, registration no. PCE18CS101 of the POORNIMA COLLEGE OF ENGINEERING Department of Computer Engineering, has submitted this project report entitled "Optimization of Railway Reservation System using Reinforcement Learning" under the supervision of Ms. Archika Jain, working as Asst. Professor in department of Computer Engineering as per the requirements of the Bachelor of Technology program of Poornima College of Engineering, Jaipur.

(Name: Dr. Surendra Kumar Yadav)

Head, Dept of Computer Engineering

(Name: Mr. Vishal Choudhary)

Project Coordinator

Dr. Mahesh Bundele
B.E., M.E., Ph.D.
Director
Poornima College of Engineering
ISI-6, RIICO Institutional Area
Sitapura, JAIPUR

DEPARTMENT CERTIFICATE

This is to certify that Ms. Nishtha Gorana, registration no. PCE18CS103, of the POORNIMA COLLEGE OF ENGINEERING, Department of Computer Engineering, has submitted this project report entitled "Optimization of Railway Reservation System using Reinforcement Learning" under the supervision of Ms. Archika Jain, working as Asst. Professor in department of Computer Engineering as per the requirements of the Bachelor of Technology program of Poornima College of Engineering, Jaipur.

(Name: Dr. Surendra Kumar Yadav)

Head, Dept of Computer Engineering

(Name: Mr. Vishal Choudhary)

Project Coordinator

Dr. Mahesh Bundele
B.E., M.E., Ph.D.
Director
Poornima College of Engineering
ISI-6, RIICO Institutional Area
Sitapura, JAIPUR

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Nabh Sharma
Nandini Singhal
Niharika Gupta
Nishtha Gorana

Dr. Mahesh Bundele
B.E., M.E., Ph.D.
Director
Poornima College of Engineering
ISI-6, RIICO Institutional Area
Sitapura, JAIPUR

CHAPTER 1
INTRODUCTION

The Indian railway is a government organization. The network is one of the largest networks in the world and is considered the lifeline of the country with almost 70,000 km long tracks, 20,000 passenger trains running on the network providing the essential services to the people on around 7000 stations spread across the country. Along with passenger services, Indian railways also provide freight services across the country with almost 8700 freight trains on average. It has a huge chunk of term budget which is provided since it require constant repair and maintenance..

In the fiscal year 2019-20, Indian Railways carried over 20 billion passengers and transported over 3 billion tonnes of freight, and generated a staggering revenue of almost 6 trillion INR. Interestingly, the freight services account for around 2/3rd of the total revenue. It is also one the biggest single organization employers. In summary, Indian railways is a gigantic economic ecosystem. Along with all the cost, it is under substantial burden to be more efficient since the demand of decent transport across the country grows enormously with each passing year. The freight traffic is too stressed; so to overcome this, dedicated freight corridors are developed which will be exclusive to freight traffic. This will drastically reduce the total freight transport time since currently preference is given to passenger trains which delays the transportation and which in turn decreases the efficiency of Indian Railways freight system. So extending railway infrastructure to satisfy the increasing demand can seems to solve the problems. But we can find certain examples across the world with high investment with no return of investment resulted in low quality across the network. Alternatively, an equally fruitful approach is to enhance the existing infrastructure using various automation techniques within the field of artificial intelligence, machine learning and internet of things. So the rest of the text consist of matters that need to be coped with along with certain logical (algorithm based) approach which can be efficiently used in the Indian economics and services context..

PROBLEM STATEMENT

Getting quicker and earliest booking in Indian Railways system is typically a challenge for passengers. The challenge is to plan an optimization using the most recent technologies which improves probability of getting a ticket, improving booking time and customer experience remarkably. Expectation from solution are:

(1) Once a passenger enters origin and destination, probable options and alternatives should be provided to him, within a predefined/limited time-frame . This might help passengers in getting confirmed seats.

(2) Additional functionality could be added to help passengers getting confirmed seats, and improve booking time.

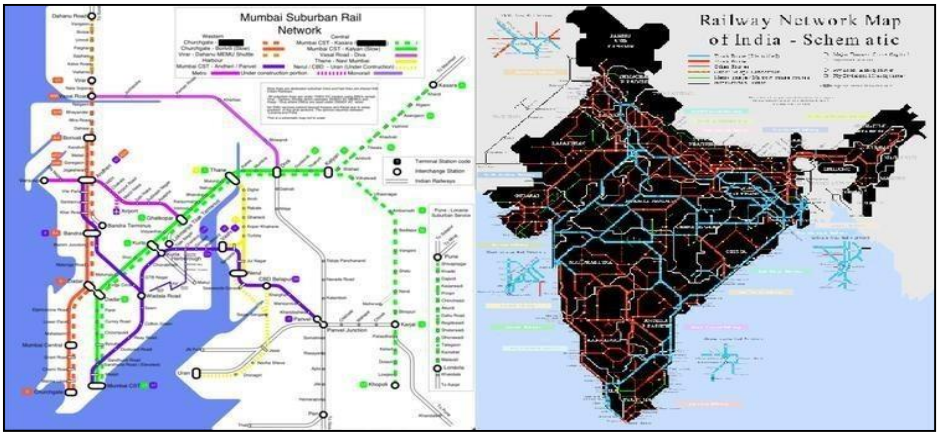
(3) Data available within the general public domain/internet are often employed by students to approach problems

2.1 POTENTIAL SOLUTIONS

Layover Routes:- The conventional booking system only shows the direct trains if available between the source and destination. So to provide more alternative routes to the passengers ,we have suggested an alternative algorithm which shows all the other feasible routes options with one or more layovers (just like air travel) in between source and destination keeping various constraints such as total duration of journey, total cost of journey along with various other constraints which can affect a passenger’s travel experience.

Early Tatkal Prediction:- Taking into account the time duration of tatkal quota booking being should out within seconds of opening. We can develop a predictive model using the previous year data of the same day which will provide us a probability of getting confirmed tickets in the tatkal slot.

2.2 SCHEDULING



Scheduling is one the most tricky problems in the railway field. Broadly , it can be categorized into 5 subproblems. The Indian railway system works theoretically as it is meant to consist of a long string of stations, which connect with each other at Major stations called the Junctions. Further, Each station contains 1 or more parallel lines, which are assigned to a particular traffic direction or might be bidirectional. The track between each pair of neighboring stations are typically called ‘sections’, which allow it to be differentiated from the tracks which are at stations

The section tracks might be bidirectional or fixed direction for the traffic. A typical section in Indian railway consists of 1 or 2 tracks, a little busy line contains more than 2 tracks which can further increase as we approach a big junction station.

Currently, these tracks are shared by freight and passenger trains, sometimes reserved for specific movement.

Datasets Used:

❖ Datasets:

- All Train Schedule
- Intermediate Stations
- Arrival Departure of Trains
- Route Rank

❖ API's:

- Train between Station i.e. Source and Destination
- Station name to Station code
- Train Live Status

CHAPTER 2
LITERATURE REVIEW

To understand the project topic, various methods can be adopted. We used various research papers for the references. At the time of the review process of those papers, we used various parameters to find the best solution approach. A summary of these review papers is presented below:

Summary Review Papers

Paper 1:

[Dr. Ram Sethu et al. 2018] has presented a paper on "A STUDY IN INDIAN RAILWAYS PASSENGERS RESERVATION SYSTEM". In this paper, the authors have explored different issues of implementing smart computing in railway systems pertaining to reservation models. He discussed the shortcomings of existing Reservation System Model along with architecture for new model implementation with UID which is UNIQUE IDENTIFICATION which would be obtained through Aadhar Card and thus removes the hassle of passengers having to enter details and payment procedure again for each booking. Thus this Proposed Model would reduce the overall turnaround time for a passenger to book a ticket which reduces the overall overhead for the servers. The only drawback in this study is it assumes that all the passenger who are booking tickets posses the facility of mobile phone at ease along which the requirement of one time registration for UID even if required ticket is booked through Station Window

Paper 2:

[Dr. Reddy et al. 2018] has presented a paper on "Enhancement of Railway Reservation System Using Internet of Things".In the present PRS system there are many passenger dilemmas to book

the ticket and railway reservation system to issue the ticket. Passengers must wait on the IRCTC website for more than 5 minutes to book a ticket. Hence present PRS is updated to smart railway reservation under the vision of India 2022. The passenger vending machine reads the passenger details that can be interacted with the railway reservation system to the UIDAI database. The result proved that passengers obtained the ticket within a fraction of a second, the reservation chart contains the passengers name with photo, and effective passenger reservations will be maintained.

Paper 3:

[Ankit Malhotra et al. 2014] has presented a paper on "Optimized Proposed Algorithm for Graph Traversal" .This paper spotlights an algorithm to traverse a directed or an undirected graph. The aim here is to hunt out a alternative method which might be invariably applied to all or any kinds of graphs the algorithm work with traversing the vertex with most outgoing edges first along with all the neighboring vertices.

If the graph is infinite and there is no solution the algorithm will diverge but within the worst case, the time complexity is analogous to Breadth First Search (BFS) i.e. the proposed algorithm possesses to think about all paths to all or any or any possible nodes .So the time complexity will nlogn in worst case.

Paper 4:

[Akshat Shrivastava et al. 2018] has presented a paper on "RAILBOT: A Railway (IRCTC) Chatbot". This paper explains the procedure to develop a Chatbot which will be providing basic information related services like PNR, seat availability, etc which in turn saves the overall time of the user to access the above information. They go to repair the bugs and include the railway complaints and booking information. The key to effective chatbot use is to make sure they enhance your customer's experience. This chatbot will benefit the passengers by saving time, memory space and internet usage.

Paper 5:

[Harshvardhan Mishra and Renuka Sharma et al. 2019] has presented a paper on “Railway Reservation and Route optimization System". Nowadays Railway is the simplest because it allows for long journeys as compared to others. this way of transportation is cost effective and appropriate for travelling. For travelling by railways there is a requirement of advanced reservation. there is a requirement to implement a system application for reservation tickets. Nowadays everyone desires to achieve the destination as fast as possible. Thus, one additional feature has been added to the prevailing reservation system and one can easily get to know which train is that the simplest option to achieve . Some sorting algorithms are employed with some circumstances. Time for all the trains has been sorted. Then by selecting this point we'll know which trains are available. The results of those methods is that, it'd show the train among all the trains at the very best of the list which is taking less time to achieve the destination. during this paper ‘Rapid Application Development process model’ has been used. By presumptuous passenger’s necessities an application has been established, thus all the necessities of passengers are often fulfilled.

Paper 6:

[Xingkui Li ET al.2019] has presented a paper on “Optimizing the Paths of Trains Formed at the loading area during a Multi-loop Rail Network”. Each loop during a multi-loop rail network consists of two segments, both of which have roughly the same conditions and mileage and are approximately symmetrical. This paper is devoted to optimizing the paths of trains formed at the loading area during a multi-loop rail network. to realize this goal, three different situations are analyzed, and two models are proposed for networks with adequate and inadequate capabilities. Two integer applied math models for optimizing the paths of trains formed at the loading area during a multi-loop rail network are proposed. The model during this paper doesn't need to determine the trail set beforehand but only must select the arc for each loop and eventually form a path by connecting each arc. a gaggle of numerical experiments with various rail loops and train flows are conducted to gauge the performance of the proposed methods. We use the Lingo solver to unravel the proposed models. For large-scale train flows and rail loops, a genetic

algorithm is supposed . This algorithm performs well in optimizing the paths of large-scale train flows formed at the loading area during a multi-loop rail network.

Paper 7:

[Hannah Bast et al. 2015] has presented paper on "Route Planning in Transportation Networks" This survey is an introduction to the state of the art within the world of practical algorithms for routing in transportation networks. This survey features a broader scope, also including schedule-based public transportation networks also as multimodal scenarios (combining schedule-based and unrestricted modes). Considers shortest path algorithms for static networks; although it focuses on methods that employment well on road networks, they're going to be applied to arbitrary graphs. Then considers the relative performance of these algorithms on real road networks, also as how they're going to affect other transportation applications. Consider algorithms for journey planning on schedule-based public transportation systems (consisting of buses, trains, and trams, for example), which is kind of different from routing in road networks. (Like time-dependent components are included publicly transport).Then consider a true multimodal scenario, which mixes schedule-based means of transportation with less restricted ones, like walking and cycling. This problem is significantly harder than its individual components, but reasonable solutions can still be found.

Paper 8:

[Ian Osband et al. 2018] has presented a paper on "Behavior Suite for Reinforcement Learning ". This paper introduces the Behavior Suite for Reinforcement Learning, or bsuite for brief . Bsuite could also be a set of carefully-designed experiments that investigate core capabilities of reinforcement learning (RL) agents with two objectives. First, to collect clear, informative and scalable problems that capture key issues within the planning of general and efficient learning algorithms. Second, to review agent behavior through their performance on these shared benchmarks.

An efficient RL agent must address three challenges simultaneously:

Generalization: be able to learn efficiently from data it collects.

Exploration: prioritize the right experience to seek out from.

Long-term consequences: consider effects beyond just one occasion step.

Paper 9:

[Mounaim Latif et al. 2016] has presented a paper on "Cross platform approach for mobile application development: A survey" during this paper a survey of cross-platform creation approaches with a stress on the MDA (Model Driven Architecture) approach because it's one of the foremost promising cross platform approaches. The codes to be written once and deployed on several platforms. Through this, time and price are reduced since the code is written to specialise in multiple devices. Developers got to maintain the model of the appliance so on lookout of the native code for each platform.

Paper 10:

[Omoniyi Ajoke et al. 2020] has presented a paper on "Design of a Modified Dijkstra's Algorithm for locating alternate routes for shortest-path problems with huge costs". The classical Dijkstra's algorithm has been widely utilized in shortest path problems. it's indeed one of the foremost referenced shortest path algorithms. However, it's observed that in certain circumstances finding shortest paths between given nodes could even be achieved at a huge cost especially where time and efforts are of great consideration. The methodology entailed creation of a document containing the nodes, edges and probabilities of moving between edges. the chances were generated employing a random number generation scheme. The contents of the document were read character by character and thus the proposed Dijkstra's algorithm. The algorithm was tested employing a 40- node graph; and thus the possible routes were generated. The findings revealed the shortest route and implies therefore that the proposed modified.

CHAPTER 3

Implementation

The Project is divided into 3 Main Modules:-

1.) Implementation of Reservation System Website

2.) Implementation of Train Search Algorithm

3.) Implementation of Reinforcement Learning Algorithm

Part 1: Implementation of Reservation System Website

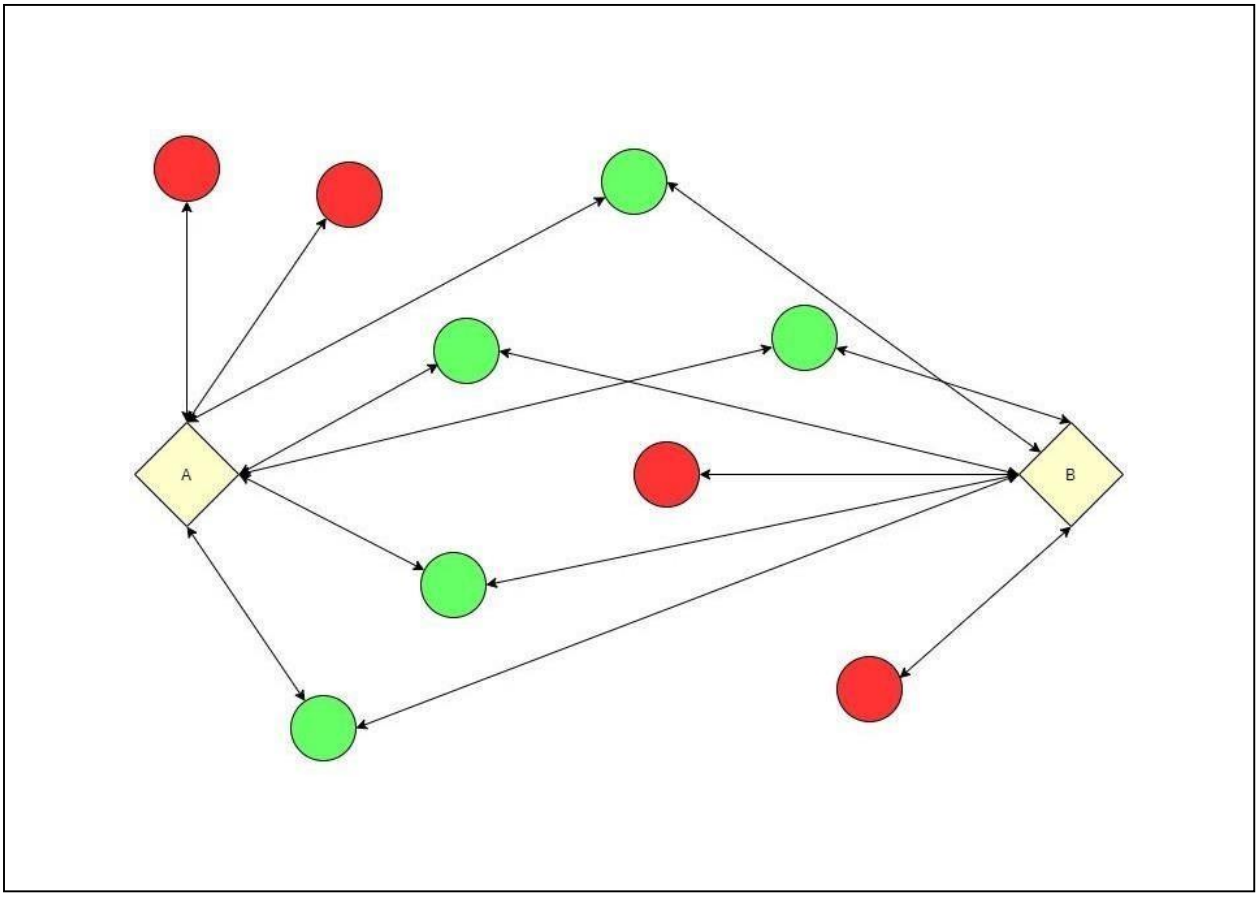
The website is based on Django Framework using Python language. A User Interface (UI) is developed to facilitate the interaction between the user and the website application. It also improves the efficiency, responsiveness and accessibility of website. We developed the UI taking the inputs i.e. Source, destination, date, train type and shift, after taking these inputs it will show the possible alternate routes to the user. We have used Django based on Python for the backend and Basic HTML, CSS, Bootstrap for the Front-end.

The UI is sleek and basic as to keep the turnaround time in between the search as minimum as possible since including animations or effects can affect the website performance which was the key aspect of the project.

Part 2: Implementation of Train Search Algorithm

The Main hurdle was to collect the respective data since Indian railway doesn't provide an official API for data collection for private usage. So we had to collect through various public gateways. After further research we stumbled upon https://indianrailapi.com/ which was a free to use platform providing data API but since it was free to use; the API provided had a really bad output efficiency. So we finally decided to store dummy data locally and proceed with our algorithm which will work on the dummy dataset. We had developed Python Script which worked as a Web Crawler using a beautiful Soup Library which helped in creation of respective data sets. Due to Covid-19 circumstances, only special Trains were working at the time of collection of data. So we have collected data of nearly 13000 railway stations and close to 250 trains which are running in real time. Since no official API was provided, some of the data had to be manually collected.

We developed an algorithm which would take source and destination from the user and provide with various possible direct as well as layover routes. The Main idea behind the algorithm is to use the Real life characteristic that all the trains run in pairs. It starts with finding all the possible stations which can be reached from the source station all over the India, considering it set A. Again, repeat the same step with finding all the possible stations which can be reached from the destination station all over the India, considering it set B. Now after we find the intersection of SET A & SET B, the resultant set provides us all the possible stations which can be considered a Layover Point irrespective of its feasibility.



Since we would be providing alternate routes to the user, the provided route is expected to be as feasible as possible. This includes feasibility factors such as **Time, Overall cost of travel and Passenger Comfort. Time** is one of the highest priority when a user is expecting alternate routes, so we have used different time factors based on the total direct journey duration between source and destination. **Overall Cost of Travel** is applied strictly as it can affect the choice of user to select a particular path. **Passenger Comfort** is also taken into account which takes woman safety in account which in turn removes all the routes which have mid – night layover suggestions and also try to accommodate the layover at the station with all the available facilities.

Part 1: Implementation of Reservation System Website

Reinforcement learning works on a set of learning problems where an agent takes actions in an environment in order to maximize some defined reward function. Since no previous data of booking and passenger reservation is provided by Indian Railway, so we had to use reinforcement learning algorithm here which work on the basic principle of reward function maximization.

So we have defined the basic reward function of (CURRENT RANK+1)/TOTAL ROUTES which changes when a user confirm the booking of one particular route from all the available routes. This influences the system to show the most ranked route as more favorable and display it topmost.

This allows the user to select the most favorable routes which also includes routes with least layovers, with day time layover when travelling with kids or ladies and also cost optimized route which is the most cost friendly to the user.

Chapter 4
Introduction to Python

Python is an object-oriented language, interpreted, dynamic and multipurpose programming language.

It was created by Guido Van Rossum in 1989. Like perl, python source code is also available under GNU general Public License (GPL).

Python is derived from many languages, including ABC, Modula-3, C, C++, Algol-68, small talk, UNIX shell and other scripted languages.

Many large companies use the python programming languages include NASA, Google, YouTube

Python Features

There are lots of features provided by python programming languages

- Python is easy to use and a high-level language. Thus, it is a programmer-friendly language.
- It is more expressive. The sense of expressiveness of the code is easily understandable.
- Python can equally on different platforms such as Windows, Linux, Unix, Macintosh etc.
- It is freely available (www.python.org). The source-code is also available. Therefore, it is open source.
- Python has a large and broad library.
- Graphical user interfaces can be developed using python.
- It can be easily integrated with languages like C, C++, JAVA etc.

3.1 Python Applications

Python have an extended wide range of applications

- Web Development :- It is extensive used in web development paradigm as it is easy to use and provide with micro-frameworks which allows fast pace development of any project
- Machine Learning and Artificial Intelligence:- It is also used in ML and AI since it provides an extended file support and have an large sized community working on to improve it
- Desktop GUI :- It is also used in desktop GUI development
- Game Development: - It also has libraries such as PySoy which are used for game development. Example: Civilization-IV, Vega Strike, etc.

3.2 Tokens:

Tokens are the smallest unit of a python program which in turn divides each line into the smallest lexical component. Each of these tokens is a subpart of a line of code.
There are following tokens in Python:

- Keywords
- Identifiers
- Literals
- Operators

3.2.1 Keywords or Reverse Words

Keywords are special pre-defined words which have specified meaning to the compiler.
List of Keywords used in Python are:

True	Try	Or	Return	Pass
Asset	For	If	From	Import
Else	Finally	Elif	Del	Except
Global	Def	Class	Continue	Break
Nonlocal		In	Not	Is
Lambda				
Raise	False	None	And	As

3.2.3. Operators

- Arithmetic Operators
- Relational Operators
- Assignment Operators
- Logical Operators
- Membership Operators
- Identity Operators
- Bitwise Operators

Arithmetic Operators:

//	Integer Divide
+	Addition
-	Subtraction
*	Multiplication
/	Division
%	return remainder
**	Exponent

Relational Operators:

<	Less than
>	Greater than
<=	Less than or equal to
>=	Greater than equal to
==	Equal to
!=	Not equal to
<>	Not equal to (similar to !=)

Logical Operators:

And	Logical AND
Or	Logical OR
Not	Logical NOT

Identity Operators:

is Returns true if identity of two operands are same, else false
is not Returns true if identity of two operands are not same, else false
Table 2.7. identity operators

3.3 Comments

- comment (single line):

Put # in the starting of the line of code
#example for this type of comment

- comment (Multiple Line):

It can be written inside triple quates.

```
"""" Here
     Is
     Multiple line comment""""
```

Assigning values to Variable
Assigning single value to multiple variables

```
x=y=z=50
print x
print y
print z
```

O/p

```
50
50
50
```

Assigning multiple values to multiple variables

```
a, b, c=5, 10, 15
print a
print b
print c
```

O/p

```
5
10
15
```

3.4 Selective Statements

The if statement is a condition checker point in a program which is executed if the condition mentioned is True or in case of False the else portion is executed if any

Simple if

```
if (conditions):
    statements
```

if with else

```
if (conditions):
    statements
else:
    statements
```

Ladder if

```
if statement:
    body
elif statements:
    body
else:
    body
```

Iteration

- While
- For

While Loop

It is also a conditional checker statement with a body of its own which in turn is executed consecutively until the condition specified is satisfied.

```
while <expression>:
    body
```

Example

```
a=10
while a>0:
    print” Value of a is”, a
    a=a-2
```

CHAPTER 5
Machine Learning

5.1 Supervised learning:

Supervised learning is the place where you've input factors (x) and a yield variable (Y) and you utilize a calculation to fit within the design capacity from the contribution to the yield.

Y = f(X)

The objective is to surmise the planning capacity so well that when you've got new info information (x) that you simply simply can foresee the yield factors (Y) for that information. It's called regulated learning on the grounds that the cycle of a calculation gained from the preparation dataset is often thought of as an educator administering the training cycle. We all know the right answers, the calculation iteratively makes forecasts on the preparation information and is adjusted by the trainer . Learning stops when the calculation accomplishes a worthy degree of execution.

5.1.1 linear regression

Linear Regression is an appealing model on the grounds that the portrayal is so straightforward. The portrayal could also be a straight condition that joins a selected arrangement of knowledge esteems (x) the answer that's that the anticipated yield for that set of data esteems (y). Therein capacity, both the info esteems (x) and thus the yield esteem are numeric. The direct condition doles out one multiplier to each information worth or segment, called a coefficient and spoken to by the capital Greek letter Beta (B). One extra coefficient is likewise added, giving the road a further level of opportunity (for example going here and there on a two-dimensional plot) and is typically called the block or the predisposition coefficient.

5.1.2 KNN

KNN are often used for both classification and regression predictive problems. However, it's more widely utilized in classification problems within the industry. to gauge any technique, we generally inspect 3 important aspects:

1. Ease to interpret output
2. Calculation time
3. Predictive Power

5.1.3 Logistic Regression

Logistic Regression could also be a directed learning arrangement calculation used to anticipate the likelihood of an objective variable. the thought of target or ward variable is dichotomous, which suggests there would be just two potential classes. In basic words, the reliant variable is double in nature having information coded together or the other 1 (represents achievement/yes) or 0 (represents disappointment/no).

Numerically, a strategic relapse model predicts P(Y=1) as a component of X. It's one of the littlest amount of complex ML calculations which can be utilized for various grouping issues, as an example , spam recognition, Diabetes expectation, malignant growth location then forth.

5.1.4 NAIVE BAYES

It is a characterization strategy enthusiastic to Bayes' theorem with a supposition of autonomy among indicators. In straightforward terms, a Naive Bayes classifier expects that the presence of a specific element during a category is random to the presence of another element.

For instance, a natural product might be viewed as an apple within the event that it's red, round, and around 3 crawls in measurement. regardless of whether these highlights depend upon one

another or upon the presence of varied highlights, these properties autonomously increase the likelihood that this organic product is an apple which is that the rationale it's mentioned as 'Gullible'

5.2 Unsupervised Machine Learning

Unsupervised learning is the place where you simply have input information (X) and no comparing yield factors. The target for unsupervised learning is to mean the essential structure or appropriation within the knowledge to become familiar with the knowledge . These are called unsupervised learning in light of the actual fact that not within the least like managed learning above there are no right answers and there is no instructor. Calculations are left to their own devices to hunt out and present the fascinating structure with regards to the knowledge .

5.3 NLP (Natural Language Processing)

Natural Language Processing, or NLP for brief , is comprehensively characterized because the programmed control of characteristic language, almost like discourse and text, by programming. The investigation of tongue preparing has been around for over 50 years and outgrew the world of etymology with the ascent of PCs.

CHAPTER 6

DATA SCIENCE

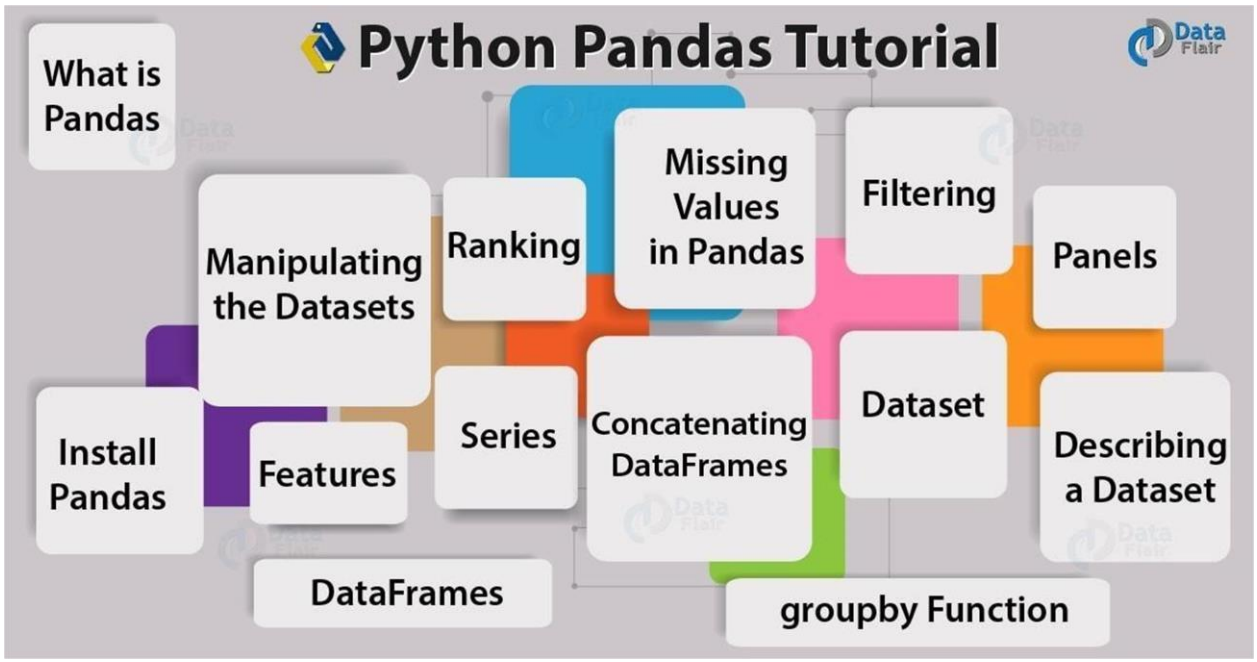
Data science is a paradigm that is combination of fundamental mathematical knowledge based on statistics along with programming skills which allows us to extract a deep understanding objectives from data. Data science usually works by application of various machine learning algorithms to various data sources i.e. texts, numbers, images, video, audio, etc. to train an artificial intelligence model which performs tasks which generally require human intelligence.

6.2 Pandas

Pandas is a python package which provides flexible ,swift data structures which are designed specifically for working with ‘labeled’ data. It is developed with an aim to act as a high level building block for doing various world data analysis in Python. Currently, the community is been working towards a broader goal of making pandas a versatile open source and a very powerful open source tool which would be available in various languages. and it is on its way towards the desired goal.

Some of the major features of Pandas are:

1. inserting or removing column in data structure
2. reshaping data sets
3. dealing with missing data
4. performing various split combine operations for data reshaping
5. large dataset label slicing operations
6. reading data from various file storage formats such as XML, CSV, etc.



Some of the major application of pandas:-

1. Economics
2. Recommendation System
3. Stock prediction
4. Advertising
5. Statistics
6. Big Data
7. Analytics
8. Data Science
9. Natural Language Processing

6.3 Matplotlib

Matplotlib may be a tremendous visualization library in Python for 2D plots of arrays. Matplotlib could also be a multi-platform data visualization library built on NumPy arrays and designed to work with the broader SciPy stack. it had been introduced by John Hunter within the year 2002. One of the simplest benefits of visualization is that it allows us visual access to large amounts of data in easily digestible visuals. Matplotlib consists of several plots like line, bar, scatter, histogram etc.

CHAPTER 7

Conclusion and Future Scope

Conquering the difficulties faced by the Indian Railways system with respect to booking, we have effectively executed the initial step of our proposed solution. This progression would give simplicity to the passengers in getting a confirmed booking when they enter the origin and destination, by giving probable options and alternatives routes to reach their desired destination.

Future Scope

- Improvements in the conventional train between station algorithms used by using specifically generated constraints by the choice of the user.
- Prediction of new train routes using the data obtained by the booking algorithms
- Improve the overall user experience at the time of booking and significantly reduce the booking time taken.